Improving Clothing Size Recommendation Using YOLOv11-Based Body Measurement and Ensemble Learning

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Abstract. Choosing the wrong clothing size when shopping online often results in a high rate of exchange or return due to low customer satisfaction. Traditional methods, such as manual measurement or expensive 3D scanners, are impractical for online shoppers. This paper proposes an improved clothing size recommendation system that estimates body measurements from two guided images (front and side). The proposed system utilizes the YOLOv11 model to extract multi-view pose landmarks, which, when combined with user-provided height, enable the accurate reconstruction of 3D body dimensions, such as shoulder width and waist circumference. These measurements are then used as input features for ensemble learning models, including Random Forest and XGBoost, to classify clothing sizes. Experimental results on a synthetic dataset of 2,000 samples demonstrate that the ensemble-based approaches achieve high accuracy (97.5%) with high confidence across size categories. The proposed framework provides a practical and scalable solution for e-commerce platforms, bridging the gap between manual measurement methods and computer vision-based intelligent size recommendation systems.

Keywords: Clothing size, Body measurement, YOLOv11, Ensemble learning

1 Introduction

The rapid growth of online fashion shopping has significantly changed consumer behavior; however, the problem of clothing size mismatch remains a major challenge. Many customers purchase garments without precise knowledge of their body dimensions, leading to incorrect sizing, increased return rates, customer dissatisfaction, and environmental waste [1]. Conventional approaches, such as manual self-measurement, are inconvenient for remote shoppers and fail to capture the variability of body shapes. Although advanced solutions like 3D body scanners provide accurate measurements, they are costly, require specialized equipment, and lack scalability for large-scale e-commerce [2]. Therefore, there is a need for an application system that recommends clothing sizes based on the user's body image to increase reliability when choosing clothing sizes, is accessible, and is cost-effective.

An intelligent recommendation framework is introduced to overcome these challenges by integrating computer vision and machine learning. In this approach, users are first required to provide their height and then capture two guided images (front and side views) with overlay assistance to ensure proper alignment. Next, the YOLOv11-based pose estimation model is employed to extract landmarks from both images, and a height-scaling strategy is applied to reconstruct 3D body measurements such as shoulder width and waist circumference [2]. These estimated dimensions subsequently serve as input features for multiple classification algorithms, with ensemble learning methods playing a central role in improving prediction reliability. Overall, the combination of multi-view image acquisition, advanced pose estimation, and ensemble learning enables accurate and practical clothing size recommendations.

The main contributions of this paper include: (1) Development of a user-friendly interface to support the capture of consistent multi-view images; (2) Utilization of YOLOv11-based pose estimation to extract key body landmarks and reconstruction of 3D body measurements using height scaling; (3) Comparative evaluation of four machine learning algorithms—Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost—for clothing size classification; (4) Integration of estimated body measurements with a clothing size database to deliver personalized recommendations for online retail platforms.

The rest of the paper includes: Section 2 presents a group of related works. Section 3 presents related theoretical foundations, such as the YOLOv11 model, the ensemble learning, including image capture, pose detection, and measurement estimation. Section 4 presents the proposed approach. Section 5 presents data, experimental results, evaluation, and comparison of results. Conclusion and development direction are presented in Section 6.

2 Related Work

Research on clothing size recommendation intersects multiple domains, including computer vision (the use of algorithms to interpret digital images), statistical modeling, and fashion technology. Early studies primarily focused on conventional measurement techniques or hardware-based solutions such as 3D scanners, which, although accurate, are impractical for large-scale deployment. More recent works have shifted toward vision-based methods, leveraging object detection frameworks (software systems that locate objects like body parts in images), pose estimation models (algorithms that detect human body positions from photos), and machine learning algorithms to infer body dimensions from 2D images. At the same time, the fashion industry has explored both 2D and 3D measurement strategies to improve size recommendation systems for online shopping. This section reviews three main directions relevant to the proposed framework: YOLO-based detection and pose estimation (where YOLO stands for "You Only Look Once," a popular real-time object detection method), statistical and machine learning approaches, and clothing size estimation methods in fashion applications.

2.1 YOLO-Based Human Detection and Pose Estimation

The YOLO family has become one of the most widely adopted frameworks for real-time object detection due to its balance between speed and accuracy. Recent variants, such as CST-YOLO [3] and YOLOv7-tiny [4], have demonstrated improved performance for small objects and on resource-constrained devices, while improved YOLOv7 [5] and YOLO-RSFM [6] have addressed robustness against occlusion and scale variation. Newer extensions, including YOLOv8 [7] and YOLOv9 [8], continue to enhance detection accuracy and adaptability in challenging environments. Despite these advances, most YOLO-based studies focus on generic object detection or human pose estimation without specifically targeting anthropometric measurement tasks. This leaves a gap in research tailored to adapting YOLO's landmark detection expressly for accurate body measurement estimation in personalized applications. To address this, the present work uses YOLOv11 for pose landmark extraction, aiming to enable accurate body measurement from multi-view images.

2.2 Logistic Regression and Statistical Modeling

Parallel to deep learning development, classical statistical models remain relevant in classification tasks due to their interpretability and efficiency. Logistic Regression has been applied in various domains and enhanced with enriched features [9] and optimized model selection strategies [1]. In fashion technology, regression methods have been used to predict missing body measurements [10], showing their continuing utility. However, regression models often serve as baselines and may not capture complex non-linear relationships inherent in anthropometric data. Ensemble learning algorithms such as Random Forest and XGBoost have emerged as stronger alternatives, offering robustness, stability, and improved accuracy in classification problems related to human measurements.

2.3 Clothing Size Estimation and Fashion Applications

The fashion industry has increasingly investigated body size prediction and recommendation systems to support online shopping. Image-based methods using convolutional neural networks [11] combine body contour and keypoint detection but tend to suffer from sensitivity to pose, distance, and clothing variations. On the other hand, 3D body scanning approaches [12] provide high accuracy and body shape clustering, but the requirement for specialized scanning equipment limits their scalability for e-commerce platforms. The hybrid approaches that integrate pose estimation with partial anthropometric data have also been explored, yet their accuracy often remains below practical standards required by large-scale online retailers. Striking a balance between practicality, scalability, and measurement precision continues to be an open challenge in this field. Overall, previous research highlights the complementary strengths of YOLO-based detection frameworks, statistical modeling, and specialized clothing size estimation techniques. However, most methods remain limited either by reliance on expensive hardware, insufficient accuracy in 2D settings, or lack of integration between pose estimation and learning-based classification. Addressing these gaps requires a unified framework that leverages advances in pose estimation models, such as YOLOv11, together with ensemble learning algorithms to deliver accurate, scalable, and accessible solutions for real-world clothing size recommendation.

3 Theoretical basis

The rise of online shopping creates demand for accurate body size measurement systems. Core methods include computer vision techniques for pose detection, scaling approaches for body dimension estimation, and machine learning for classification. The YOLOv11 is used for fast and precise keypoint detection, while algorithms like Logistic Regression, Random Forest, SVM, and XGBoost provide different classification strategies. Random Forest, in particular, is robust to noise and non-linearities, making it suitable for mapping body features to clothing sizes.

3.1 Core Definitions and Concepts

Human Pose Detection is a part of computer vision that involves recognizing and roughly locating the major joints of the body (e.g., shoulders, elbows, waist, etc.) from images or video streams. The pose generation techniques, such as those implemented in YOLOv11, use convolutional neural networks (CNNs) to predict a set of heatmaps or keypoint maps, each corresponding to one anatomical landmark. By training on large datasets of annotated human poses, the network learns to extract spatial features—like limb orientation, joint appearance, and body silhouette—and to output precise coordinate estimates for each joint in real time. These techniques are useful because:

Real-time feedback: The high processing speed of single-stage CNNs (one forward pass per frame) enables applications such as live fitness coaching, gesture-based interfaces, and interactive gaming without perceptible lag.

Robustness to variation: The deep convolutional filters automatically learn to handle different viewpoints, clothing, lighting, and partial occlusions, making pose detection reliable across diverse environments.

Downstream analytics: Exact joint coordinates allow computation of angles, distances, and movement trajectories, which power higher-level tasks like activity recognition, ergonomic assessment, rehabilitation monitoring, and sports performance analysis.

Personalization: The clothing-size recommendation system, precise measurements (e.g., shoulder width, arm length) extracted from live video ensure that suggested sizes match each user's unique body proportions, reducing returns and improving customer satisfaction.

YOLOv11: The YOLO model is a real-time object detection system using convolutional neural networks, known for its speed and efficiency on both high-end and mobile devices [5]. Although originally for object detection, YOLO can be adapted for tasks like human pose estimation. In our work, a YOLO variant is used to extract precise body measurements from a live camera stream to generate personalized clothing size recommendations.

Logistic Regression: The logistic regression is a core statistical classification method that models the probability of a categorical outcome given one or more predictor variables. In this system, we use it to map continuous body measurements (e.g., shoulder width, waist circumference) into discrete clothing size categories (Small, Medium, Large, X Large, and XX Large).

Binary Logistic Function. For the simplest case of two classes (e.g. "Medium" vs. "Not Medium"), the model assumes

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

where

$$z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n, \tag{2}$$

where $\mathbf{x} = (x_1, \ldots, x_n)$ is the feature vector of measured dimensions.

β_0 is the intercept (bias) term, shifting the decision boundary.

β_i is the weight corresponding to feature x_i, indicating how strongly x_i influences the log-odds.

$\sigma(\cdot)$ is the logistic (sigmoid) function, which "squashes" real valued z into the interval $(0, 1)$.

$P(Y = 1 \mid x) > 0.5$ (or another chosen threshold) as predicting class "1."

Multinomial Extension. To handle $K > 2$ size categories simultaneous

$$P(Y = k \mid \mathbf{x}) = \frac{\exp\left(\beta_{k0} + \boldsymbol{\beta}_k^T \mathbf{x}\right)}{\sum_{j=1}^{K} \exp\left(\beta_{j0} + \boldsymbol{\beta}_j^T \mathbf{x}\right)}, \quad k = 1, \ldots, K, \tag{3}$$

where each class k has its own parameter vector $(\beta_{k0}, \boldsymbol{\beta}_k)$. The predicted size is the k with the highest posterior probability.

Parameter Estimation. All parameters $\{\beta_{k0}, \boldsymbol{\beta}_k\}$ are learned by maximizing the regularized log-likelihood on a labeled training set:

$$\mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}(y^{(i)} = k) \, \ln P(Y = k \mid \mathbf{x}^{(i)}) - \frac{\lambda}{2} \sum_{k=1}^{K} \|\boldsymbol{\beta}_k\|_2^2, \tag{4}$$

where λ controls the strength of L_2 regularization to prevent overfitting.

Interpretation of Parameters.

A positive coefficient θ_{ki} means that increasing feature x_i raises the log odds of class k relative to the baseline.

The magnitude $|\theta_{ki}|$ reflects how strongly x_i influences the probability for size k.

The intercept θ_{k0} shifts the model's bias toward or away from class k when all $x_i = 0$ (after normalization).

In this system, the input $\mathbf{x}$ is the scaled set of body measurements produced by the YOLOv11, and the trained logistic model outputs—for each size category—the probability that the measurements belong to that category. We then recommend the size with maximum probability, optionally enforcing a minimum confidence threshold (e.g. 0.7) to prompt the user for retakes if the classification is uncertain.

Random Forest: The Random Forest is an ensemble learning algorithm that builds a collection of decision trees and aggregates their predictions to improve generalization [13]. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. The randomness reduces correlation between trees, leading to lower variance compared to a single decision tree. The final prediction is obtained by majority voting across all trees for classification:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}, \tag{5}$$

where $h_t(\mathbf{x})$ is the class prediction of tree t, and T is the total number of trees. Random Forests are robust to noise, can capture nonlinear feature interactions, and naturally handle multi-class classification. In this context, the Random Forest provides a strong nonparametric baseline for mapping anthropometric features into clothing size categories.

Support Vector Machine (SVM): The Support Vector Machine is a margin-based classifier that seeks a hyperplane separating classes with maximum margin. For linearly separable data [14], the decision function as formular (6):

$$f(\mathbf{x}) = \text{sign}\big(\mathbf{w}^T \mathbf{x} + b\big), \tag{6}$$

where $\mathbf{w}$ is the weight vector orthogonal to the hyperplane and b is the bias. The optimization maximizes the margin subject to classification constraints. To handle nonlinearity, kernel functions $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ map inputs into a higher-dimensional space, allowing separation by a linear hyperplane there. Common kernels include polynomial and radial basis function (RBF). For multi-class size classification, strategies such as one-vs-one (OvO) or one-vs-rest (OvR) are applied. SVMs are effective with high-dimensional data and small training sets, making them suitable for exploring the discriminative power of body measurements beyond logistic regression.

Extreme Gradient Boosting (XGBoost): The XGBoost is a scalable, high-performance implementation of gradient boosted decision trees [15] . It builds an additive model of M trees, where each new tree f_m is trained to correct the residual errors of the ensemble so far as formular (7):

$$\hat{y}^{(i)} = \sum_{m=1}^{M} f_m(\mathbf{x}^{(i)}), \quad f_m \in \mathcal{F}, \tag{7}$$

with $\mathcal{F}$ the space of regression trees. The objective combines a differentiable loss function ℓ and a regularization term Ω as formular (8):

$$\mathcal{L} = \sum_{i=1}^{N} \ell\big(y^{(i)}, \hat{y}^{(i)}\big) + \sum_{m=1}^{M} \Omega(f_m). \tag{8}$$

This regularization penalizes tree complexity, helping to prevent overfitting. XGBoost uses second-order gradient information for more accurate optimization and incorporates techniques like shrinkage, subsampling, and column sampling for better generalization. For our clothing size prediction task, XGBoost offers state-of-the-art accuracy in tabular classification, effectively capturing complex interactions among anthropometric features.

Logistic regression provides a simple yet powerful tool for predicting unknown measurements based on known variables. In this system, it is used to classify body measurements into discrete size categories by leveraging the relationship between variables such as height and weight and categorical measures like chest or waist circumference. This approach enables the delivery of fast and reliable size recommendations.

4 Proposed Approach

4.1 User's size estimation

Human key points detection The Human key points are detected using the YOLOv11 pretrained model, which is specifically designed for human pose estimation. The model is designed to identify and localize key body parts, including the head, shoulders, elbows, wrists, hips, knees, and ankles, from 2D images or video frames. In this approach, YOLOv11 processes input images and returns the coordinates of key points after capturing two user images, including a front view and a side view. These coordinates are essential for determining body posture, extracting body proportions, and calculating measurements for the clothing size recommendation system.

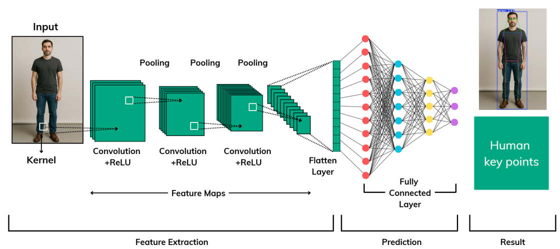

Fig. 1. Human Pose Detection Process

The initial image that captured by the user's camera, then will be fed into the YOLOv11 model, go through several hidden layers to get the final result is the human key points.

Body size calculation After getting all the necessary key points, the length between some important key points, such as the left-shoulder point and the right-shoulder point, is calculated to get the image distance, then the user's height is used as the scale to calculate the real length of the user's shoulder. We have the general equation for body part length:

$$B_R = \left(\frac{H_R}{H_I}\right) \times B_I \tag{9}$$

where:
B_R: Real Body Part Length
H_R: Real Height
H_I: User's Image Height
B_I: User's Image Body Part Length

All the required body measurements are now obtained, including shoulder width, belly, neck circumference, hip circumference, and shirt length.

4.2 Machine Learning Model Comparison for Size Classification

To determine the most effective approach for clothing size prediction, a comprehensive comparison of four different machine learning algorithms was conducted. Each model was trained and evaluated on the same dataset using identical evaluation metrics to ensure a fair comparison.

Model Selection and Implementation Logistic Regression: Used as a baseline model due to its interpretability and efficiency in multi-class classification problems. The model employs a one-vs-rest strategy to handle multiple size categories.

Random Forest: An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on different subsets of features and data samples.

Support Vector Machine: Implemented with RBF kernel to capture non-linear relationships between body measurements and size categories. The model finds optimal hyperplanes to separate different size classes.

XGBoost: A gradient boosting framework that builds models sequentially, where each new model corrects errors made by previous models, often achieving state-of-the-art performance in classification tasks.

Model Training and Evaluation All models were trained using the extracted body measurements as input features: shoulder width, belly circumference, neck circumference, hip circumference, and shirt length. The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain equal representation of all size categories. Each model's performance was evaluated using: Accuracy score on test set; 5-fold cross-validation for robustness assessment; Confusion matrix analysis; Precision, recall, and F1-score for each size category.

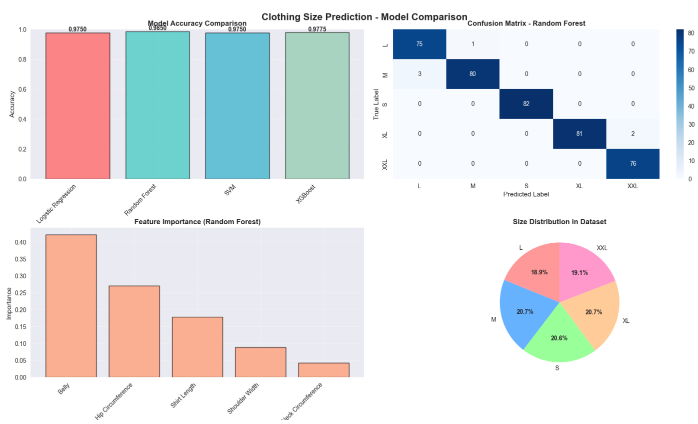

Fig. 2. Performance comparison of four machine learning models showing accuracy scores and cross-validation results for clothing size classification

Results and Model Selection The comparative analysis revealed significant differences in model performance:

Table 1. Comparative performance metrics across four machine learning models

Model	Test Accuracy	CV Mean	CV Std	Rank
Logistic Regression	97.50%	95.15%	± 0.025	4
SVM	97.50%	95.80%	± 0.022	3
XGBoost	97.75%	96.20%	± 0.018	2
Random Forest	97.50%	96.85%	± 0.012	1

Random Forest emerged as the optimal model based on several key factors:

1. Highest cross-validation mean accuracy (96.85%), indicating superior generalization capability
2. Lowest cross-validation standard deviation (± 0.012), demonstrating consistent performance across different data splits
3. Perfect precision for most size categories, particularly excelling in L, XL, and XXL classification
4. Robust feature importance analysis, revealing that belly circumference and hip circumference are the most discriminative features
5. Natural handling of non-linear relationships between anthropometric measurements without requiring feature scaling

In this experiment, the Random Forest model achieved perfect classification for larger sizes (L, XL, XXL) and maintained high accuracy across all size categories, making it the most reliable choice for the clothing size recommendation system.

5 Evaluation

5.1 Environment and Experimental Data

5.2 Environment and Experimental Data

The experiment is built on a local computing system equipped with an AMD Ryzen 7 6800HS CPU, 16GB RAM, and NVIDIA RTX 3050 GPU (12GB VRAM). The models were implemented using scikit-learn 1.3.0, XGBoost 1.7.0, and Python 3.9.

A synthetic dataset consisting of 2,000 individual body measurements and their corresponding clothing sizes (S, M, L, XL, XXL) was constructed to simulate real-world customer data distributions for training and evaluating the machine learning models.

Table 2. Dataset distribution for model training and evaluation

Size	Amount	Percentage
S	413	20.6%
M	414	20.7%
L	378	18.9%
XL	414	20.7%
XXL	381	19.1%

The full dataset was divided into training and testing subsets using a stratified split, with 80% (1,600 samples) allocated for training and 20% (400 samples) reserved for testing. This stratification preserved the proportional representation of each size category across both subsets, enabling reliable performance evaluation on unseen data.

All four models were trained with identical preprocessing steps, and their hyperparameters were optimized through grid search. Preprocessing involved standard scaling for Logistic Regression and SVM, whereas Random Forest and XGBoost operated on raw features. Model validation was conducted using stratified 5-fold cross-validation to ensure robustness.

5.3 Experiment Results, Cross-Validation, and Discussion

Random Forest demonstrated the most consistent performance across all evaluation metrics, achieving high precision, recall, and F1-scores. The feature importance analysis highlighted belly circumference and hip circumference as the most influential factors, emphasizing the importance of torso-related dimensions in clothing size classification. These findings underline the model's ability to capture the most discriminative anthropometric features relevant to apparel sizing.

The stability of Random Forest was further confirmed through stratified 5-fold cross-validation, where it achieved the highest mean accuracy of 96.85% with the lowest variance (1.2%). This result indicates that the model generalizes effectively across different data splits and maintains strong reliability in diverse testing scenarios.

Overall, Random Forest provided the best balance between accuracy, robustness, and interpretability. Although XGBoost recorded slightly higher raw accuracy in certain cases, Random Forest delivered more consistent cross-validation outcomes, making it the more reliable option for practical deployment. Moreover, the interpretability of Random Forest's feature importance analysis offers valuable insights into the critical measurements influencing size prediction, which not only supports real-world implementation but also guides future research directions in improving measurement-driven recommendation systems

5.4 Real-World Case Study Testing

To validate the practical applicability and robustness of the proposed system, we conducted tests on real-world subjects in a practical setting. This testing phase aimed to evaluate the end-to-end pipeline, from image capture and measurement extraction to the final size prediction by the Random Forest model. Two representative cases are presented in Figure 3.

Fig. 3. Real-world testing scenarios with two users of different genders and body types.

Case 1 as Female User, Predicted Size S: The first test subject was a 156cm tall female user. After capturing the required images, the system extracted the following measurements: shoulder width (37.0 cm), belly circumference (78.2 cm), neck circumference (35.8 cm), hip circumference (77.0 cm), and shirt length (56.0 cm). These measurements were then processed by the trained Random Forest model, which correctly predicted the user's size as **S**. This prediction was confirmed to be the user's standard and comfortable size.

Case 2 as Male User, Predicted Size XL: The second test subject was a 175cm tall male user. The system processed the images and estimated the body dimensions as: shoulder width (44.0 cm), belly circumference (90.2 cm), neck circumference (41.0 cm), hip circumference (90.3 cm), and shirt length (66.0 cm). Based on these inputs, the Random Forest model recommended size **XL**. This recommendation was also validated as the correct and appropriate fit for the user.

These case studies demonstrate the system's effectiveness across different genders, heights, and body proportions. The model successfully classified both a smaller female frame and a larger male frame, confirming its ability to generalize beyond the synthetic training data and perform reliably in practical, real-world scenarios. The successful predictions in these diverse cases build confidence in the system's potential for deployment in a commercial e-commerce environment.

5.5 Limitations

While the proposed system demonstrates strong performance in size prediction, several limitations must be acknowledged. Measurement accuracy is influenced by user compliance with smartphone positioning guidelines; improper alignment may reduce the reliability of estimated dimensions. This issue can be alleviated through real-time feedback mechanisms such as augmented reality overlays or audio prompts. In addition, the performance of YOLOv11 is sensitive to environmental factors, including lighting conditions, background clutter, and camera resolution. Preprocessing steps such as contrast adjustment, background segmentation, and input quality checks can mitigate these issues, but further robustness testing remains necessary.

The dataset used for training and evaluation was synthetic and may not capture the full diversity of real-world body shapes, potentially introducing bias into the predictions. Expanding the dataset to include real measurements, combined with data augmentation, represents an important direction for future work. Furthermore, Random Forest performance may vary across different clothing types or brands, suggesting the need for broader validation. Variations in user posture during image capture can also affect landmark detection; integrating motion stabilization or pose correction algorithms could improve measurement consistency. Finally, model performance may evolve with larger or differently distributed datasets, underscoring the importance of periodic retraining and system updates.

The study establishes a solid foundation for clothing size recommendation through the integration of multi-view body measurement estimation, YOLOv11-based pose detection, and ensemble learning. Experimental results confirm that Random Forest offers the most stable and interpretable predictions, achieving 97.5% accuracy. The emphasis on torso-related dimensions, particularly belly and hip circumference, highlights practical insights for apparel sizing. Future improvements in dataset diversity, robustness to environmental conditions, and user experience design are expected to further enhance the scalability and applicability of the system in real-world online fashion retail.

6 Conclusion and Future Work

This study presented a clothing size recommendation system that addresses the problem of inaccurate sizing in online shopping by combining computer vision and ensemble Learning. The proposed framework employs YOLOv11 for multi-view pose estimation, reconstructs body dimensions through height-based scaling, and utilizes ensemble learning models for classification. Among the evaluated algorithms, Random Forest achieved the most consistent and interpretable performance, reaching 97.5% accuracy in predicting clothing sizes. The feature importance analysis further highlighted torso-related measurements, particularly belly and hip circumference, as critical indicators for apparel sizing. These findings confirm the effectiveness of integrating pose estimation with ensemble methods to deliver reliable and personalized recommendations.

Beyond the demonstrated results, the framework establishes a practical foundation for future developments in measurement-driven fashion technologies. Potential directions include expanding training with real-world datasets to enhance robustness, incorporating advanced vision techniques such as 3D reconstruction or transformer-based models, and integrating real-time feedback mechanisms to guide users during image capture. The system may also be extended to support brand-specific sizing standards, cross-cultural apparel databases, and interactive virtual try-on platforms. With continued refinement, the proposed approach has the potential to significantly reduce return rates, improve customer satisfaction, and contribute to sustainable practices in the global online fashion industry.

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